

Amazon S3 Storage Classes,

Replication & Lifecycle Management

week 3| part 2

How AWS stores, protects, copies, archives, and automatically manages data as its usage changes over time.

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Why S3 Data Management Matters

Definition S3 data management means choosing how objects should be stored, protected, copied, retained, and eventually archived or deleted.

Availability

How quickly must users or applications retrieve the object?

Protection

Does the workload require another copy in another Region or AWS account?

Cost

Can older or rarely accessed data move to a lower-cost storage class?

Retention

How long must the data remain available before archive or deletion?

ML example: recent training data may be accessed every day, while older datasets may be retained only for auditing. Keeping both in the same storage class can waste money.

PRESENTATION EXPLANATION

Explain that S3 design is not only about uploading files. A good architecture connects access frequency, protection, recovery, compliance, and cost to a specific storage decision.

Amazon S3 Replication: The Core Idea

Definition S3 Replication automatically copies eligible objects from a source bucket to a destination bucket according to a replication rule.

CR
R

Cross-Region Replication

- Destination bucket is in another AWS Region.
- Useful for disaster recovery, compliance, and regional access.
- Creates geographic separation from the source data.

SR
R

Same-Region Replication

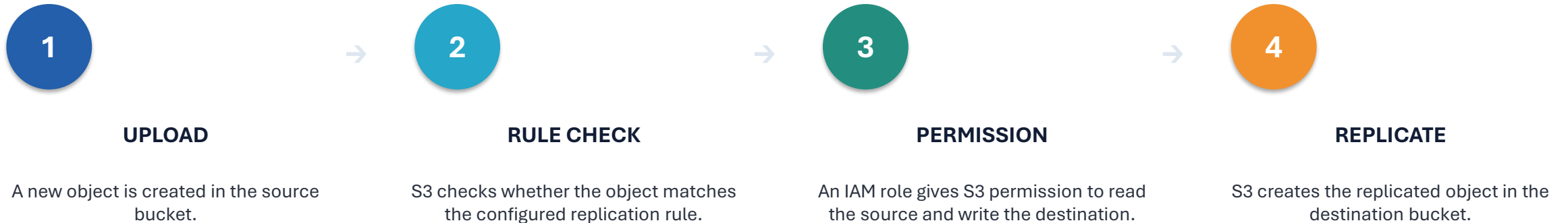
- Destination bucket stays in the same AWS Region.
- Useful for log aggregation and environment separation.
- Can also support copies across different AWS accounts.

IMPORTANT AWS POINT: Versioning must be enabled on both source and destination buckets.

PRESENTATION EXPLANATION

Use a simple backup example: CRR protects against a regional problem, while SRR is more useful when the copy is needed inside the same Region for organizational or account separation.

How S3 Replication Works



New vs. existing objects After replication is enabled, new matching objects replicate automatically. Existing objects are not automatically included just because the rule was created later.

S3 Batch Replication can be used for existing objects and objects that previously failed replication.

PRESENTATION EXPLANATION

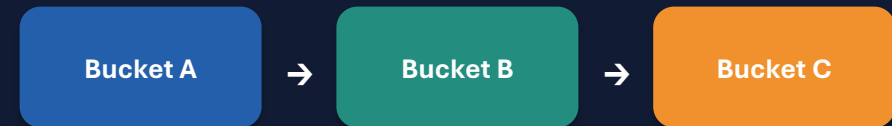
If a bucket already contains 100,000 ML images before replication is enabled, those older images need Batch Replication if the organization also wants them copied.

Replication Rules You Should Remember

DELETE BEHAVIOR

- Delete markers can optionally be replicated from source to destination.
- A deletion that specifies a particular version ID is not replicated.
- This behavior helps prevent a malicious or accidental permanent delete from automatically removing the same version in the destination.

NO REPLICATION CHAINING



If A replicates to B and B replicates to C, an object originally created in A is not automatically chained onward to C.

PRESENTATION EXPLANATION

These rules matter for recovery design. Replication is controlled and intentional; it should not be treated as an unlimited automatic backup chain.

Amazon S3 Storage Classes

Definition

Storage classes are different S3 storage options designed for different access frequencies, availability needs, retrieval expectations, and costs.

S3 Standard

Frequently accessed data

Standard-IA

Infrequent but rapid access

One Zone-IA

Infrequent + recreatable

Glacier Instant

Archive + millisecond retrieval

Glacier Flexible

Archive + flexible retrieval

Glacier Deep Archive

Very rare long-term access

S3 Intelligent-Tiering is another option that can automatically adjust access tiers when object access patterns change.

PRESENTATION EXPLANATION

Do not memorize storage classes only by name. Ask: How often is the data used? How quickly must it return? Can it be recreated? How long will it be stored?

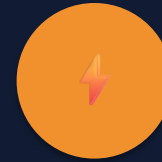
Durability vs. Availability



DURABILITY

Question: “Will my object remain intact?”

Amazon S3 is designed for 99.999999999% durability — commonly called 11 nines. The course notes this durability across the storage classes in this section.



AVAILABILITY

Question: “Can I access my object right now?”

Availability measures how readily the service can be accessed and varies by storage class. The course lists S3 Standard at 99.99% availability.

PRESENTATION EXPLANATION

A file may still exist safely but be temporarily unavailable. That is why durability and availability are separate measurements and should not be used as if they mean the same thing.

S3 Standard, Standard-IA & One Zone-IA

S3 STANDARD

- 99.99% availability
- Frequently accessed data
- Low latency + high throughput
- Designed to sustain two concurrent facility failures
- Examples: big-data analytics, mobile/gaming apps, content distribution

STANDARD-IA

- 99.9% availability
- Less frequently accessed
- Rapid access when needed
- Lower storage cost than Standard
- Examples: disaster recovery and backups

ONE ZONE-IA

- 99.5% availability
- Stores data in one Availability Zone
- Lower-cost infrequent access
- Data can be lost if that AZ is destroyed
- Best for secondary copies or recreatable data

PRESENTATION EXPLANATION

The cheapest option is not automatically the best option. For example, One Zone-IA is appropriate only when a single-AZ copy is acceptable or the data can be recreated.

S3 Glacier Storage Classes

Purpose Glacier classes are low-cost S3 storage options intended mainly for archiving and backup, where data is valuable but accessed much less frequently.

GLACIER INSTANT RETRIEVAL

- Millisecond retrieval
- Good for data accessed about once per quarter
- 90-day minimum storage duration

Machine Learning archive examples

- Historical training datasets
- Previous model artifacts
- Experiment outputs
- Compliance or audit records
- Old application and inference logs

KEY IDEA

Archiving does not mean deleting. It means retaining important data in a storage class that better matches rare access.

PRESENTATION EXPLANATION

ML projects continuously create datasets, model versions, logs, and experiments. Glacier helps retain older information without treating it like active working data.

S3 Lifecycle Rules: Automatic Data Movement

Definition An S3 Lifecycle configuration automatically performs actions on objects when they reach specified ages or match selected conditions.

TRANSITION ACTIONS



EXPIRATION ACTIONS

- Delete access logs after a defined period.
- Remove old object versions.
- Delete incomplete multipart uploads.
- Target objects by prefix or object tags.

PRESENTATION EXPLANATION

Lifecycle rules remove manual work. They are especially useful when millions of objects follow the same business rule, such as moving old datasets to archive storage after a fixed period.

Lifecycle Scenario: Profile Images & Thumbnails

ORIGINAL PROFILE IMAGES

Requirement Immediate retrieval is needed during the first 60 days. After that, the user can tolerate slower retrieval.

Design Store the source images in S3 Standard, then transition them to Glacier after 60 days.

S3 Standard → 60 days → Glacier

THUMBNAIL IMAGES

Requirement Thumbnails are temporary and can easily be recreated from the original image.

Design Use One Zone-IA and configure a lifecycle expiration action to delete thumbnails after 60 days.

One Zone-IA → 60 days → Delete

PRESENTATION EXPLANATION

This scenario shows that different objects in the same application can require different storage strategies. Preserve valuable source data; remove inexpensive data that can be regenerated.

Lifecycle Scenario: Recovering Deleted Objects

Business rule Deleted S3 objects must be recoverable immediately for 30 days. After that, and for up to 365 days, recovery within 48 hours is acceptable.

1

ENABLE VERSIONING

A delete marker hides the current object instead of immediately destroying all versions.

2

NONCURRENT → STANDARD-IA

Keep noncurrent versions in a class suitable for infrequent but rapid recovery.

3

NONCURRENT → DEEP ARCHIVE

Later transition old noncurrent versions to a long-term archival class.

PRESENTATION EXPLANATION

Versioning and lifecycle management work together. Versioning preserves recoverable copies, while lifecycle rules reduce the cost of keeping those older versions for longer periods.

S3 Analytics – Storage Class Analysis

Definition

S3 Analytics – Storage Class Analysis studies access patterns to help decide when objects may be good candidates for transition between S3 Standard and Standard-IA.

WHY USE IT?

Instead of guessing that data becomes inactive after 30 or 60 days, teams can observe actual usage patterns before creating or improving lifecycle rules.

1

Updated daily

Analysis reports are refreshed each day.

2

24–48 hours

Initial analysis can take time to appear.

3

Standard + Standard-IA

Recommendations apply to these classes.

4

Not One Zone-IA / Glacier

The course notes those are not supported for this analysis.

PRESENTATION EXPLANATION

For an ML team, actual access behavior may show that researchers continue using datasets much longer than expected. Analytics can support a more realistic lifecycle policy.

Choosing the Right S3 Feature

Need a copy in another Region

Cross-Region Replication

Frequently accessed data

S3 Standard

Recreatable infrequent data

One Zone-IA

Data should move as it ages

Lifecycle Rules

Need a copy in the same Region

Same-Region Replication

Infrequent + rapid retrieval

Standard-IA

Long-term archive

Glacier storage classes

Unsure when to transition

S3 Analytics

PRESENTATION EXPLANATION

AWS questions often describe a requirement instead of naming the feature. Look for clues such as another Region, infrequent access, archive, automatic transition, or recoverable deleted versions.

Key Takeaways

- 1 Replication** Creates controlled copies of eligible S3 objects for regional, account, recovery, or organizational requirements.
- 2 Storage Classes** Match object access frequency and retrieval requirements to an appropriate storage option.
- 3 Durability vs Availability** Durability is about keeping the object intact; availability is about being able to access it when requested.
- 4 Lifecycle Rules** Automatically transition or expire objects as they age or match conditions.
- 5 S3 Analytics** Uses access-pattern analysis to support better Standard-to-Standard-IA lifecycle decisions.

PRESENTATION EXPLANATION

Close by connecting the topics together rather than memorizing them separately. The goal is to recognize the workload requirement and select the AWS feature that best matches it.